LAB 1:Complete Lab-Focused Teaching Package

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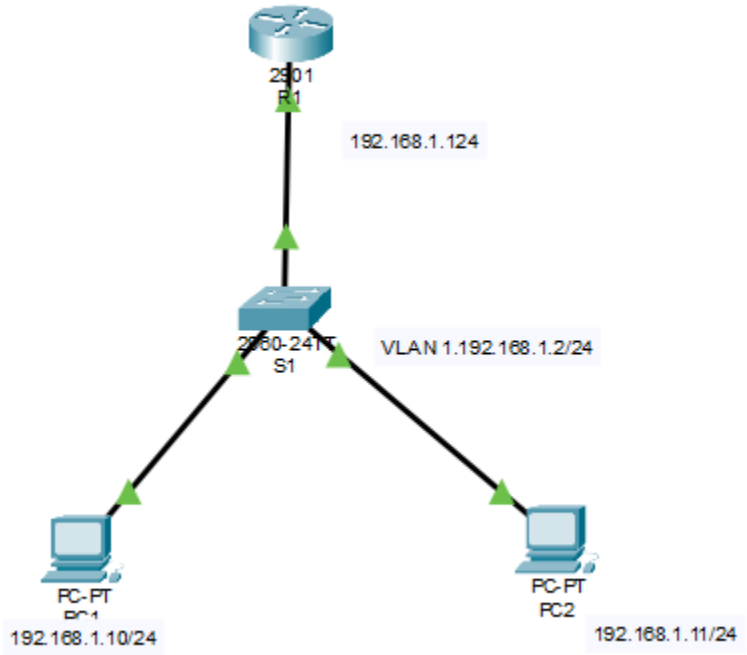
รับผิดชอบ Part 5,Worksheet 1-3(Leader)

Lab 1: Complete Lab-Focused Teaching Package

📊 TOPOLOGY DIAGRAM



📋 ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Default Gateway	MAC (Dynamic)
R1	G0/0	192.168.1.1	255.255.255.0	N/A	0001.9643.AC01
S1	VLAN 1	192.168.1.2	255.255.255.0	192.168.1.1	0002.1745.8B02
PC1	NIC	192.168.1.10	255.255.255.0	192.168.1.1	0050.7966.6803
PC2	NIC	192.168.1.11	255.255.255.0	192.168.1.1	0090.2B14.7C04

Network: 192.168.1.0/24
Usable Host Range: 192.168.1.1 - 192.168.1.254
Broadcast: 192.168.1.255

PART 2: CONNECTIVITY VERIFICATION

Ping pc1 > pc2,router,switch

Command Prompt

```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping pc2 > pc1,router,switch

```
Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping router > pc1,pc2,switch

```
Router>ping 192.168.1.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

Router>ping 192.168.1.11

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.11, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router>ping 192.168.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.2, timeout is 2 seconds:
..!!!
Success rate is 60 percent (3/5), round-trip min/avg/max = 0/0/0 ms

Router>
```

ARP Table

-PC1

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           0001.9643.ac01        dynamic
  192.168.1.2           0010.11ec.c352        dynamic
  192.168.1.11          0090.2b14.7c04        dynamic
```

-Pc2

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           0001.9643.ac01        dynamic
  192.168.1.2           0010.11ec.c352        dynamic
  192.168.1.10          0050.7966.6803        dynamic
```

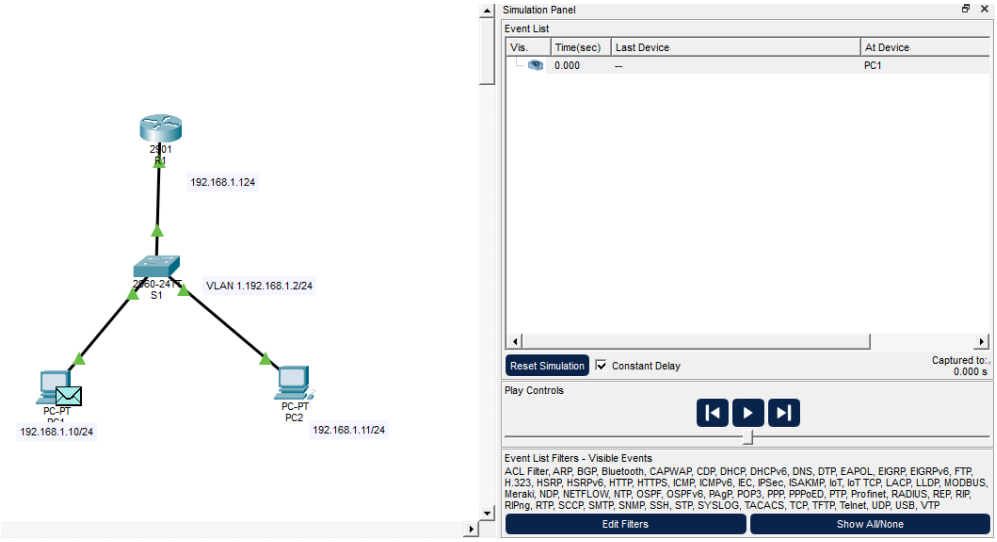
-router

```
Router>enable
Router#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.1.1            -    0001.9643.AC01  ARPA   GigabitEthernet0/0
Internet  192.168.1.2            3    0010.11EC.C352  ARPA   GigabitEthernet0/0
Internet  192.168.1.10           4    0050.7966.6803  ARPA   GigabitEthernet0/0
Internet  192.168.1.11           7    0090.2B14.7C04  ARPA   GigabitEthernet0/0
```

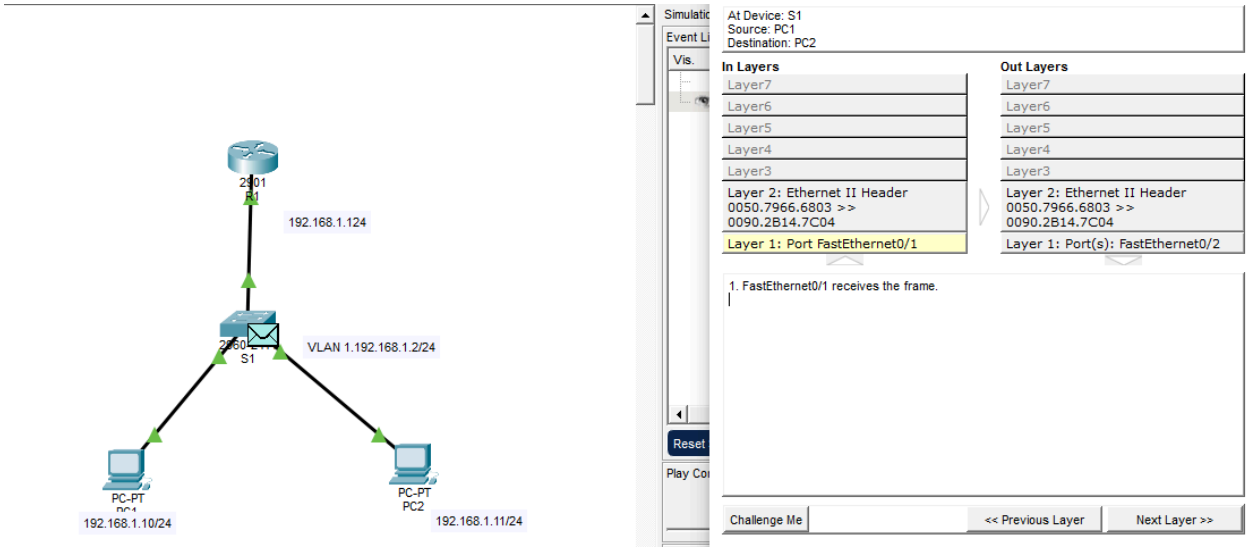
-switch

```
Switch#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.9643.ac01    DYNAMIC     Gig0/1
   1    0050.7966.6803    DYNAMIC     Fa0/1
   1    0090.2b14.7c04    DYNAMIC     Fa0/2
```

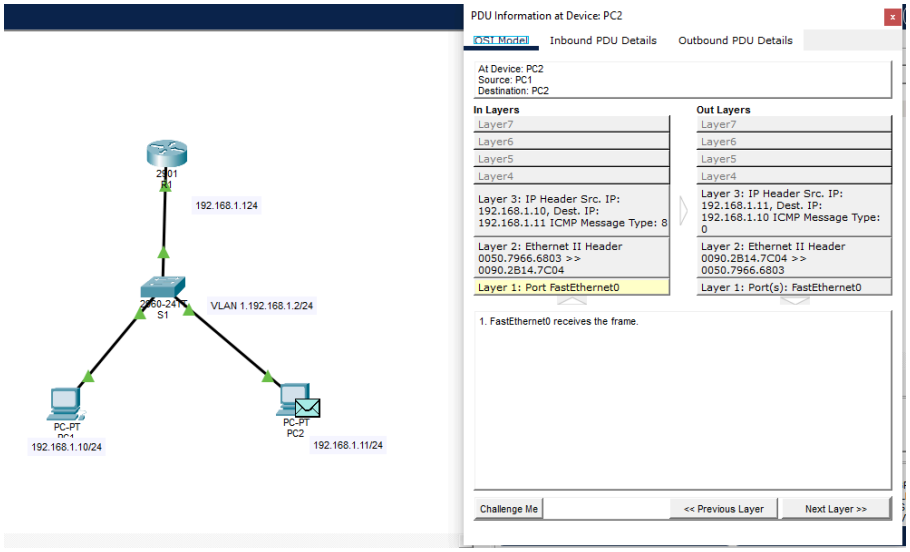
PART 3: ENCAPSULATION ANALYSIS (25 minutes)



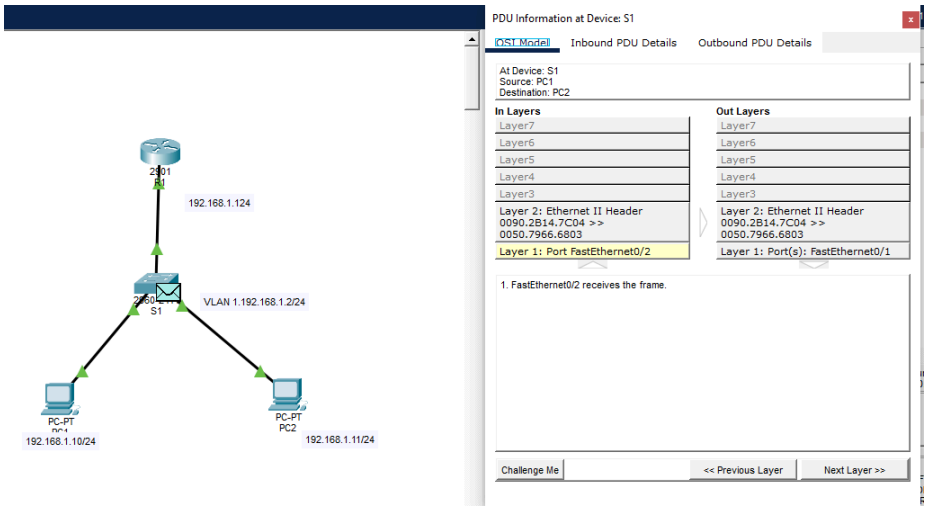
Step 6: Packet Tracer Simulation Mode



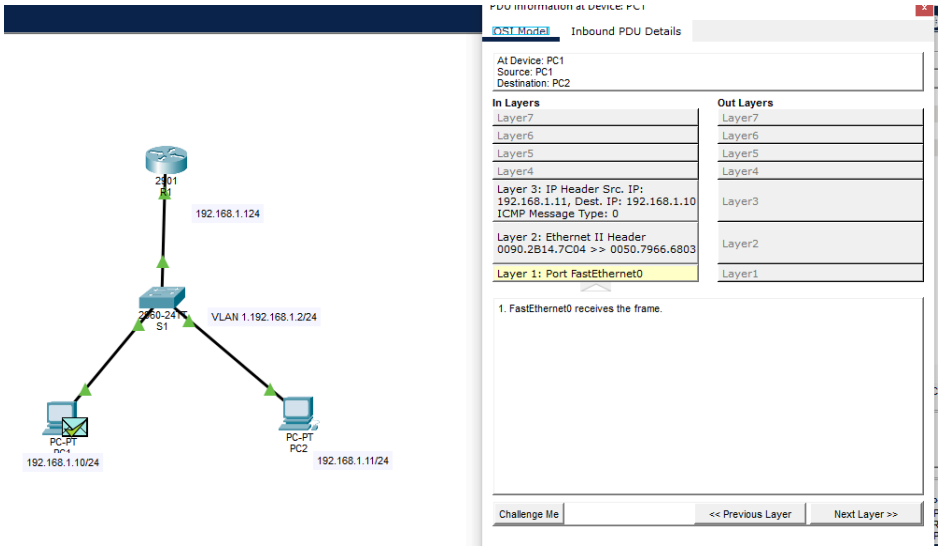
1. Enter **Simulation Mode** in Packet Tracer



2. Create a **PDU** from PC1 to PC2 (ICMP Echo)



3. **Slow the simulation** and observe each step:



Observation Points:

- **Layer 2:** Ethernet frame construction (Source/Dest MAC)
- **Layer 3:** IP header creation (Source/Dest IP)
- **ARP Process:** How PC1 resolves PC2's MAC address
- **Switch Forwarding:** How S1 uses MAC table
- **Encapsulation/Decapsulation** at each hop

Step 7: Protocol Header Documentation

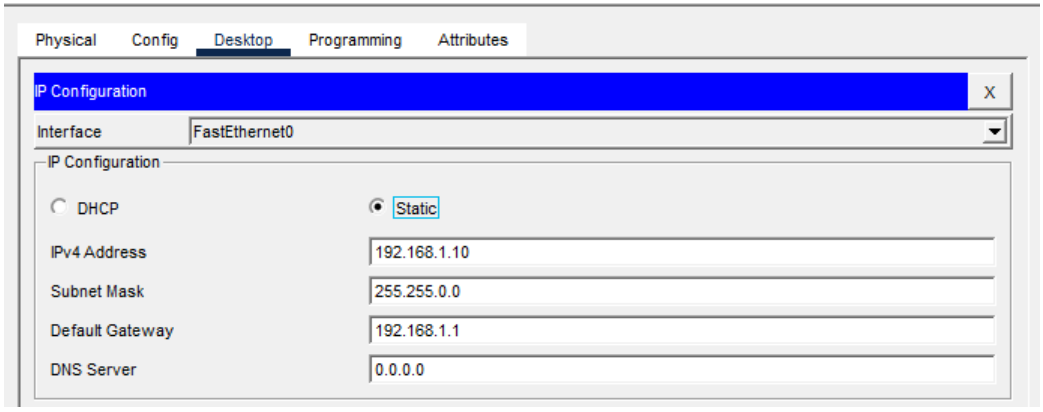
Create this table in your lab report:

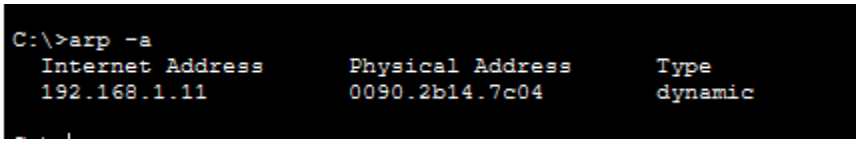
Step	Protocol	OSI Layer	TCP/IP Layer	Key Field Observed
1	ARP Request	Layer 2	Network Access	Target IP: 192.168.1.11
2	ARP Reply	Layer 2	Network Access	Sender MAC: [PC2's MAC]
3	ICMP Echo	Layer 3	Internet	Type: 8, Code: 0
4	Ethernet	Layer 2	Network Access	Frame Check Sequence
5	ICMP Reply	Layer 3	Internet	Type: 0, Code: 0

PART 4: TROUBLESHOOTING CHALLENGE (25 minutes)

Step 8: Intentional Break-Fix Scenarios

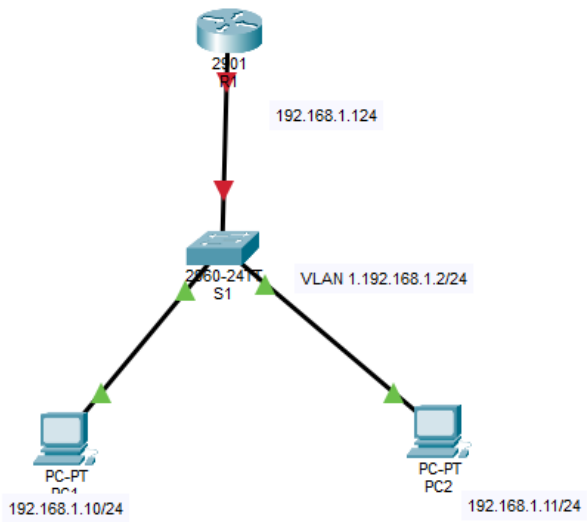
Scenario A: Incorrect Subnet Mask






```
None
! On PC1, change to wrong subnet mask:
PC1> ipconfig /ip 192.168.1.10 255.255.0.0 192.168.1.1
! Attempt ping to PC2 - Observe failure
! Use 'arp -a' to diagnose
! Fix: Restore correct mask 255.255.255.0
```

Scenario B: Router Interface Shutdown



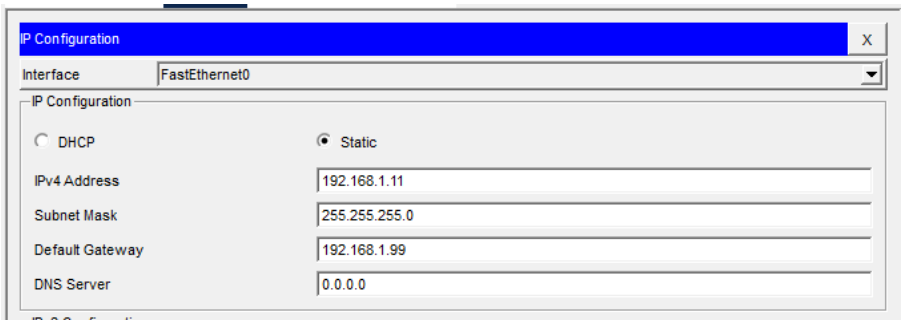

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

```
C:\>tracert 192.168.1.1

Tracing route to 192.168.1.1 over a maximum of 30 hops:

  1   *        *        *        Request timed out.
  2   *        *        *        Request timed out.
  3   *        *        *        Request timed out.
  4   *        *        *        Request timed out.
  5   *        *        *        Request timed out.
  6   *        *        *        Request timed out.
  7   *        *        *        Request timed out.
  8   *        *        *        Request timed out.
  9   *        *        *        Request timed out.
  10   *        *        *        Request timed out.
  11   *        *        *        Request timed out.
  12   *        *        *        Request timed out.
  13   *        *        *        Request timed out.
  14   *        *        *        Request timed out.
  15   *        *        *        Request timed out.
  16   *        *        *        Request timed out.
  17   *        *        *        Request timed out.
  18   *        *        *        Request timed out.
  19   *        *        *        Request timed out.
  20   *        *        *        Request timed out.
  21   *        *        *        Request timed out.
  22   *        *        *        Request timed out.
  23   *        *        *        Request timed out.
  24   *        *        *        Request timed out.
  25   *        *        *        Request timed out.
  26   *        *        *        Request timed out.
  27   *        *        *        Request timed out.
  28   *        *        *        Request timed out.
  29   *        *        *        Request timed out.
  30   *        *        *        Request timed out.

Trace complete.
```

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#
```

Scenario C: Wrong Default Gateway

```
None
! On PC2:
PC2> ipconfig /dg 192.168.1.99
! Attempt ping from PC2 to R1
! Use 'route print' (Windows) to verify gateway
! Fix: Restore to 192.168.1.1
```



PC2 -> R1

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

PART 5: ADVANCED OBSERVATION

```
C:\>ping 192.168.1.11 -n 5

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
Switch#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.9643.ac01    DYNAMIC     Gig0/1
   1    0050.7966.6803    DYNAMIC     Fa0/1
   1    0090.2b14.7c04    DYNAMIC     Fa0/2
Switch#
```

Application data generation	พิมพ์คำสั่ง `ping 192.168.1.11` บน PC1	ICMP Echo
TCP segment creation	-	ICMP
IP packet encapsulation	PC1 ใส่ IP Header ครอบ ICMP	ICMP
Ethernet framing	นำข้อมูลหาปลายทาง	Ethernet frame
Physical transmission	Frame ถูกส่งเป็นสัญญาณไฟฟ้าในสาย LAN	-

📝 LAB DELIVERABLES

Worksheet 1: Configuration Verification

Checkpoint	Command Used	Expected Result	Actual Result	Pass/Fail
R1 Interface Status	`show ip int brief`	G0/0 UP/UP	G0/0 UP/UP G0/1 DOWN/DOWN	ผ่าน
S1 VLAN Config	`show vlan brief`	VLAN 1 Active	VLAN1 Active	ผ่าน
PC1 Connectivity	`ping 192.168.1.11`	Success (4 replies)	4 replies	ผ่าน
ARP Resolution	`arp -a` on PC1	2+ entries	3 entries	ผ่าน

Checkpoint	Command Used	Expected Result	Actual Result	Pass/Fail
MAC Learning	`show mac address-table` on S1	2 dynamic entries	3 entries	ผ่าน

Worksheet 2: Protocol Analysis

Capture this data during simulation:

1. **ARP Request Frame:**

    - Source MAC: 0000.0C6A.9165
    - Destination MAC:192.168.1.10
    - ARP Opcode: 1

2. **ICMP Echo Packet:**

    - Time to Live (TTL): 255
    - Protocol Field: 1(Value = 1)

3. **Encapsulation Chain:** Data → _TCP/UDP_ Header → _IP__ Header → _Ethernet__ Header → Bits

Worksheet 3: Troubleshooting Journal

Problem Injected	Symptoms Observed	Diagnostic Commands Used	Root Cause	Solution
Wrong Subnet Mask	ปิงได้	`ping`, `arp -a`	Subnet mask ค่าที่เปลี่ยนมีค่ากว้างกว่าค่าเดิม ระบบจึงไม่พัง	เปลี่ยนกลับ
Interface Shutdown	ปิงไปR1,PC1ไม่ได้	`show ip int brief`	Router DOWN	no shutdown
Incorrect Gateway	ปกติ	`ipconfig`, `tracert`	ปกติ	เปลี่ยนกลับ

🔍 VERIFICATION COMMANDS CHECKLIST

Router (R1):

- ☑ `show running-config`
- ☑ `show ip interface brief`
- ☑ `show arp`
- ☑ `show ip route`

Switch (S1):

- ☑ `show running-config`
- ☑ `show vlan brief`
- ☑ `show mac address-table`
- ☑ `show interfaces status`

End Devices:

- ☑ `ipconfig /all` (or equivalent)
- ☑ `arp -a`
- ☑ `ping` to all devices
- ☑ `tracert` (if multi-hop)

🎓 ASSESSMENT QUESTIONS

1. When PC1 pings PC2, which protocol resolves the MAC address before ICMP operates?

   ..ARP เพราะต้องรู้ MAC Address ก่อนส่งแพ็กเก็ต IP...

2. What would happen if you changed the subnet mask on PC1 to 255.255.255.128?

   …ยัง ping ได้ใน LAN เดียวกัน..

3. Why does the switch not need an IP address for basic frame forwarding?

   ..เพราะ Switch ทำงานที่ Layer 2 ใช้ MAC Address..

4. Which OSI layers are involved when R1 forwards a packet between interfaces?
   ..OSI Layer 3…

5. What is the purpose of the TTL field in the IP header you observed?

…ป้องกันแพ็กเก็ตวนลูปในเครือข่าย…